

HIPS

The pelvic and the femur bones form the skeletal structure in the hip region of the body. The head of the femur bone fits into the acetabular fossa, a socket on the pelvis, to form the hip joint. This ball-and-socket joint allows the widest range of motion in the body. The movements of this joint include flexion, extension, abduction, adduction, and internal and external rotation of the hip. Surrounding the hip joints are several large and strong muscle groups, making possible the major movements of the lower extremities that are necessary for our daily activities.

Multiple muscles as well as several important ligaments surrounding the hip joint provide strong support. The ligamentum teres ligament connects the head of the femur and the acetabular notch of the pelvis to keep them together. The iliofemoral, ischiofemoral, and pubofemoral ligaments give extra support so that the head of the femur stays in the acetabular fossa in a firm, snug, and tight formation during all daily activities. The acetabular labrum runs along the rim of the acetabular fossa to deepen the hip cavity, thus giving additional support to the hip joint. All these structures combine to protect the hip joint and make it quite strong and able to withstand the demands of constant muscular movements.

All but two of the muscles of the hip (figure 4.1) run between the pelvic bones and the thigh bone (femur). The two exceptions are the psoas major and piriformis, which run between the lower vertebral column and the femur. The muscles that move the hip joint are some of the largest muscles (adductor magnus and gluteus maximus) in the body as well as some of the smallest (gemellus superior and inferior). The anterior (front) muscles—the psoas major, iliacus, rectus femoris, and sartorius—flex the hip and are used during walking to swing the leg forward. The posterior (back) muscles—the gluteus maximus, biceps femoris, semimembranosus, and semitendinosus—provide the backward swing for walking. A group of large muscles (adductor brevis, adductor magnus, adductor longus, gracilis, and pectineus) on the medial (inside) thigh keep the legs centered under the body. A group of small muscles (gluteus medius, gluteus minimus, piriformis, gemellus superior, gemellus inferior, obturator internus, obturator externus, quadratus femoris, and tensor fasciae latae) on the lateral (outside) thigh splay the legs to the side. Another group that makes up more than 75 percent of the hip muscles is the external hip rotators, consisting of the gluteus maximus, gluteus medius, gluteus minimus, piriformis, gemellus superior, obturator internus, gemellus inferior, obturator externus, quadratus femoris, psoas major, iliacus, rectus femoris, sartorius, adductor brevis, adductor magnus, adductor longus, and pectineus.

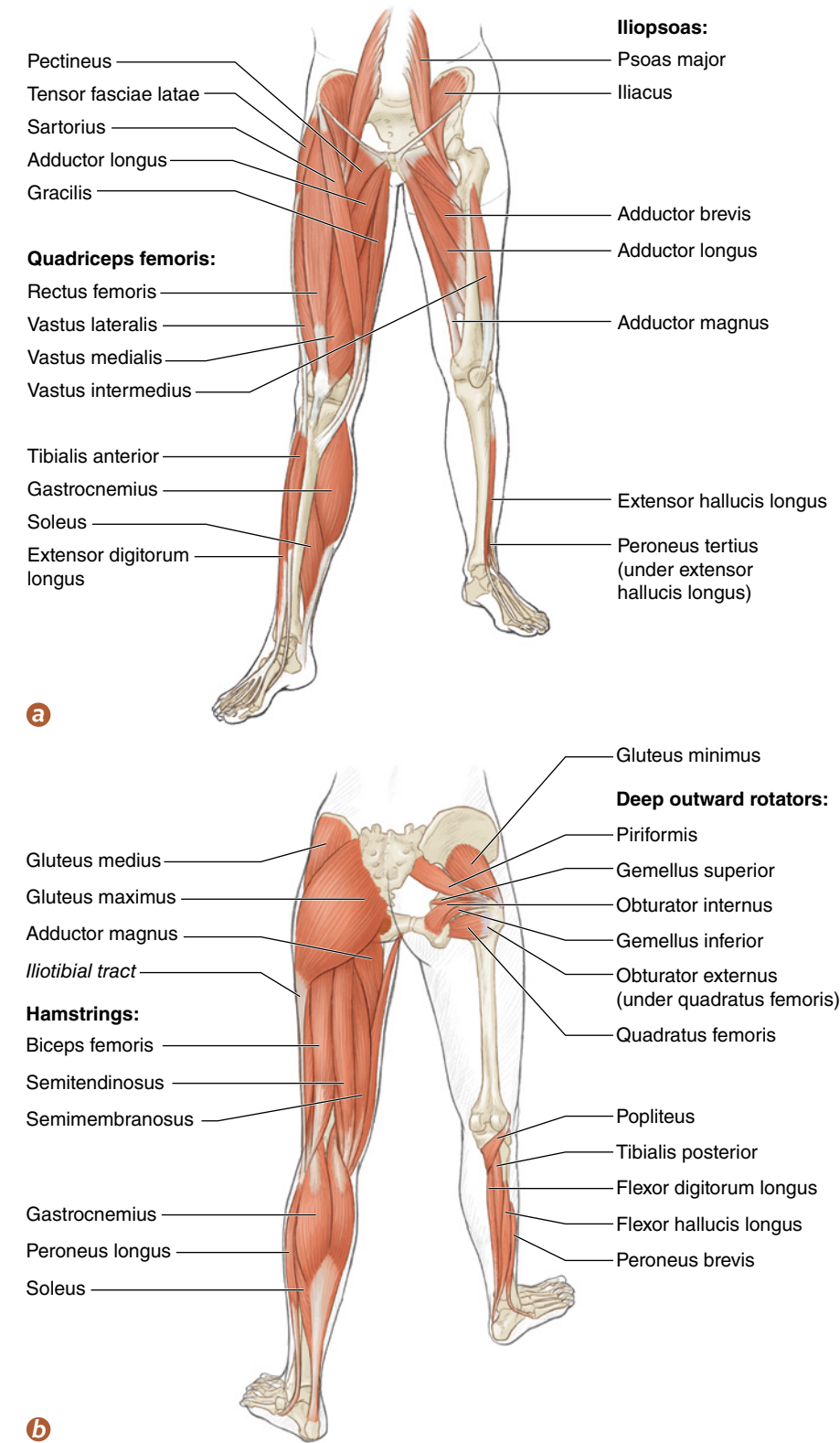


FIGURE 4.1 Muscles of the lower extremities: (a) anterior; (b) posterior.

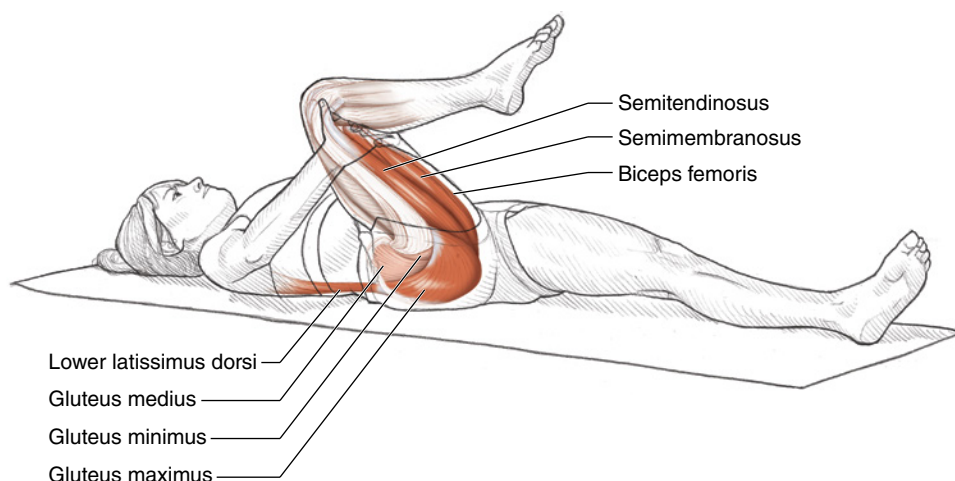
The range of motion, or the degree of freedom to move the hip, depends on several factors, including bony structure; muscle strength; stiffness of muscle tissue, tendons, and ligaments; and anatomical restrictions. For hip flexion, the range of motion is limited by hip flexor strength, stiffness of the hamstring muscles, and contact of the leg with the abdomen. Extension is influenced by hip extensor strength and the stiffness of both the hip flexors and the ligaments surrounding the ball-and-socket joint. Hip abduction is limited not only by the strength of the abductors and stiffness of the adductors but also by the stiffness of the pubofemoral and iliofemoral ligaments and bony contact of the femoral neck and acetabular rim. On the other hand, hip adduction is restricted by the strength of the adductors and stiffness of the abductors as well as the stiffness of the iliofemoral and capitate (or ligamentum teres femoris) ligaments. Besides muscle strength of the agonist muscles and stiffness of the antagonist muscles, internal rotation movement is restrained by the iliofemoral and ischiofemoral ligaments, while external rotation is restrained by tension in the iliofemoral ligament.

Flexibility has more to do with overall body function than previously thought. For instance, diminished flexibility is one indicator of an aging body. Decreased physical activity also results in decreased flexibility. As people age and decrease their physical activity, they must keep stretching muscle groups in order to maintain mobility and range of motion in the joints. The hip region is located in the middle of the body, so problems in this area tend to radiate and affect many other parts of the body. You can reduce and even prevent many hip problems by paying more attention to strength and joint flexibility.

For instance, pain in the hip or buttocks area is often associated with poor hip flexibility. This is especially true after running or hiking along steep inclines or declines or even on slanted surfaces. Hip pain that occurs one or two days after activity that uses the hip external rotator muscles extensively is caused by damage to both the muscle and the connective tissues in and around the muscle. Unfortunately, the hip external rotator muscles are small and usually weak and as such are not strengthened during typical strength-training activities. Therefore, stretching these muscles before and after the activity may help decrease this soreness and increase their strength. In addition, the hip external rotator muscles are the least-stretched muscles of the lower body, probably because these muscle groups are also the most difficult to stretch. We all tend to ignore those places in the body where we often find the most problems. On the bright side, it is easy to concentrate more on stretching those stiff and sore muscle groups.

The hip stretches in this book are grouped according to which muscle groups are being stretched. In addition, they are listed and described in order from the easiest to the most difficult. Those who are new to a stretching program tend to be less flexible and should begin with the easiest level of stretches. Progression to a more difficult stretch in this program should be made when you feel confident you are able to advance to the next level. For detailed instructions, refer to the information on stretching programs in chapter 10.

HIP AND BACK EXTENSOR STRETCH



Execution

1. Lie on your back on a comfortable surface.
2. Bend the right knee, and bring it toward the chest.
3. While keeping the left leg flat, grasp the right knee with both hands, and pull it down toward the chest as far as possible.
4. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right gluteus maximus, right erector spinae (iliocostalis, longissimus, spinalis), right lower latissimus dorsi, right hamstring (semitendinosus, semimembranosus, biceps femoris)

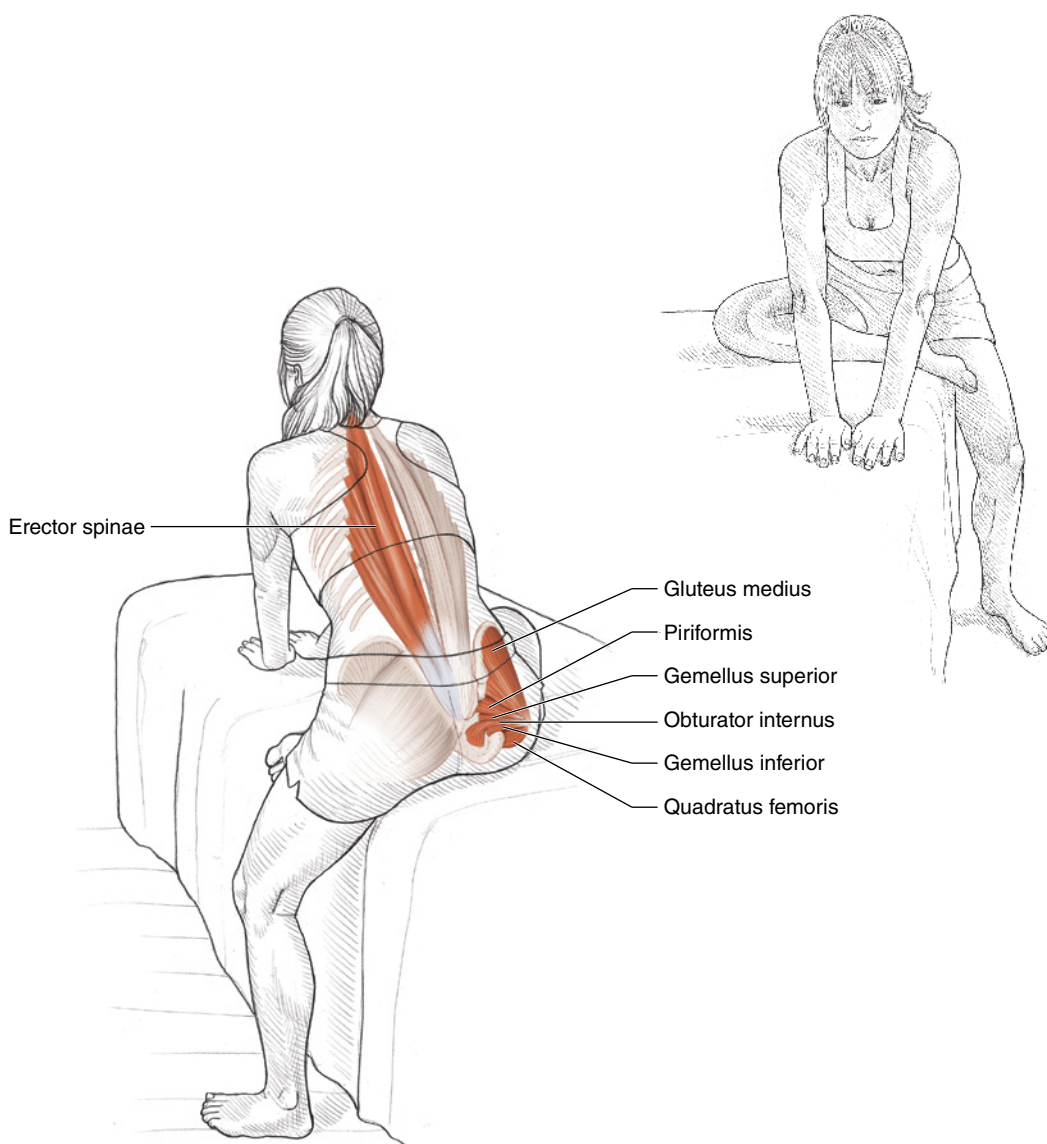
Less-stretched muscles: Right gluteus medius, right gluteus minimus

Stretch Notes

This is another helpful and effective stretch for people who suffer from lower-back, pelvic, or hip pain. Pain in the pelvic region is often a result of muscular soreness, and when muscles are sore, they often feel stiff as well. A person with this condition has a tendency to limit the range of motion of the affected muscles in order to avoid pain. Therefore, normal daily activities can be significantly affected depending on the severity of the pain. Rather than avoiding movement, a person experiencing this condition should specifically try to move and stretch the injured muscles. Performing the hip and back extensor stretch will provide increased flexibility and strength to these muscle groups, which in turn will help lessen the likelihood (or severity) of future injury.

For warm-up purposes, it is recommended that you use both legs simultaneously at first. Once warmed up, bring one knee up to the chest at a time. In addition, pulling the knee up toward the armpit will maximize the effectiveness of this stretch.

BEGINNER SEATED HIP EXTERNAL ROTATOR STRETCH



Execution

1. Sit on a bed or couch.
2. Rotate the right leg at the hip, and pull the right foot in to rest flat against the left inner thigh as close as possible to the pelvic area. The outside of the lower right leg should rest as flat as possible on the surface of the bed or couch.
3. Bend the trunk over toward the right (bent) knee as far as possible until you start to feel a slight stretch (light pain). Keep the left knee down, if possible, as you bend over.
4. As you bend over, reach out with your arms over the right foot.
5. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles on right side: Gluteus maximus, gluteus medius, gluteus minimus, piriformis, gemellus superior, gemellus inferior, obturator externus, obturator internus, quadratus femoris

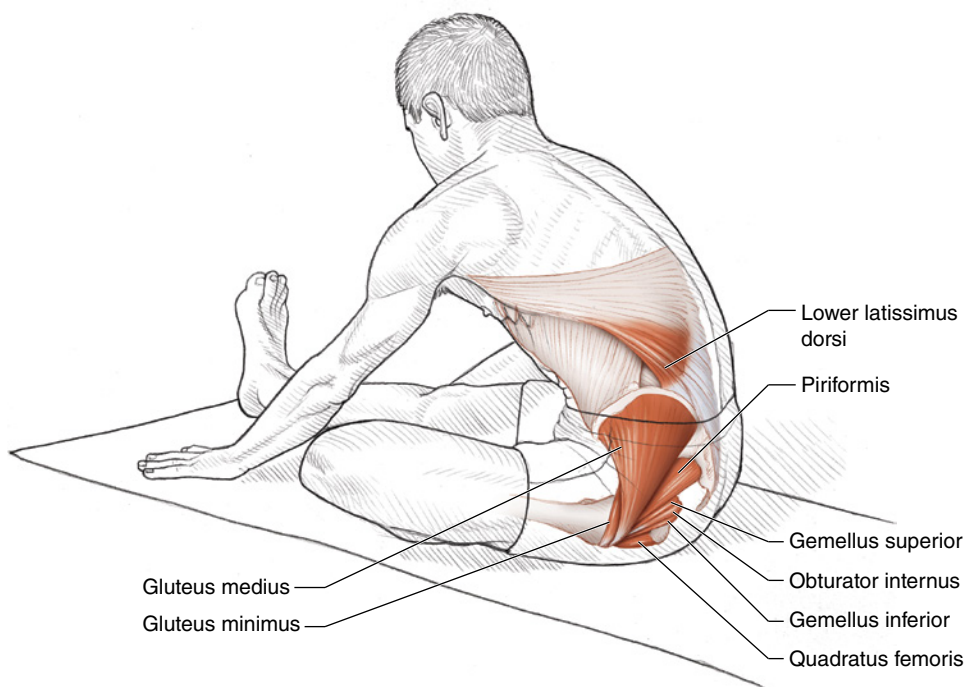
Most-stretched muscles on left side: Erector spinae (iliocostalis, longissimus, spinalis), lower latissimus dorsi

Stretch Notes

This stretch is a lowest-stress version of the hip external rotator stretches and as such is the best stretch to use at first. The small external rotator muscles of the hip are located on the outer back side of the hip, under the gluteus maximus muscle. If you feel minor tightness or soreness here, especially after walking, running activities, or climbing, use this low-intensity stretch to relieve the stress put on these muscles during these activities. You use these muscles whenever the hip rotates outward, such as when walking and running. If the external rotator muscles are not strong or flexible enough, they can become sore and tight very easily.

This particular stretch can be easily done while sitting on a couch or bed, and it is one of the easiest stretches to execute for these muscles groups. Doing this stretching exercise in a sitting position with the right leg up flat, bent 90 degree or less, on the couch surface and the left leg hanging down is a relaxing position. If you are less flexible or beginning a stretching program, it would perhaps be better to start this stretch with your right knee bent to an angle of more than a 90 degrees and then gradually work on bending the knee more as your flexibility improves. Remember to bend the trunk forward from the hip. It is also beneficial to keep the back straight; do not curl or hunch the back while performing the stretch.

INTERMEDIATE SEATED HIP EXTERNAL ROTATOR AND EXTENSOR STRETCH



Execution

1. Sit with the right leg extended straight out in front. Bend the left knee and place the left foot flat against the right inner thigh as close as possible to the pelvic area. Place the hands on the floor next to the thighs.
2. Keeping the trunk straight, bend the trunk forward from the hip joint over toward the right (straight) knee as far as possible until you start feeling a slight stretch (light pain). Keep the right knee down on the floor if possible as you bend over. Reach out with your arms toward the right foot.
3. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles on left side: Gluteus medius, gluteus minimus, piriformis, gemellus superior, gemellus inferior, obturator externus, obturator internus, quadratus femoris, erector spinae (iliocostalis, longissimus, spinalis), lower latissimus dorsi

Most-stretched muscles on right side: Hamstring (semitendinosus, semimembranosus, biceps femoris), gluteus maximus, gastrocnemius

Less-stretched muscles on the right side: Soleus, plantaris

Stretch Notes

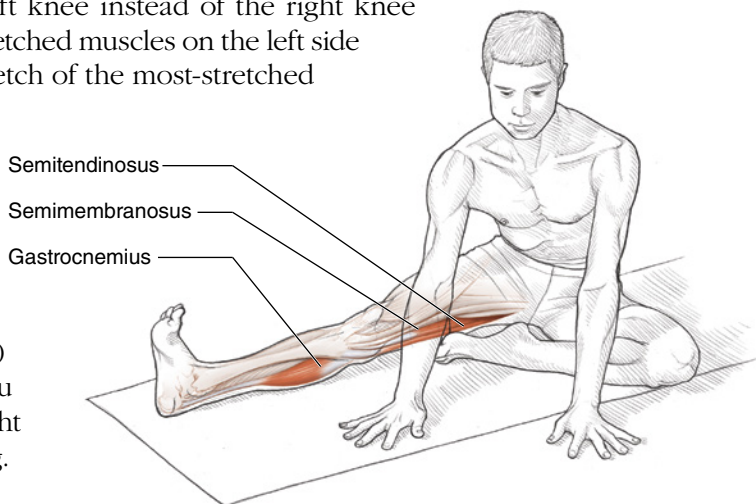
The hip external rotator muscles are commonly neglected in stretching routines. Overuse of these muscles in activities such as basketball, soccer, and hockey can lead

to soreness, tightness, and even injuries to this area. In addition, poor flexibility usually leads to lower-quality performance. Participants do a lot of stepping sideways, using a lot of these muscles whenever the hip rotates outward. Using this stretch regularly will build flexibility and strength.

VARIATION

Intermediate Seated Hip Extensor and External Rotator Stretch

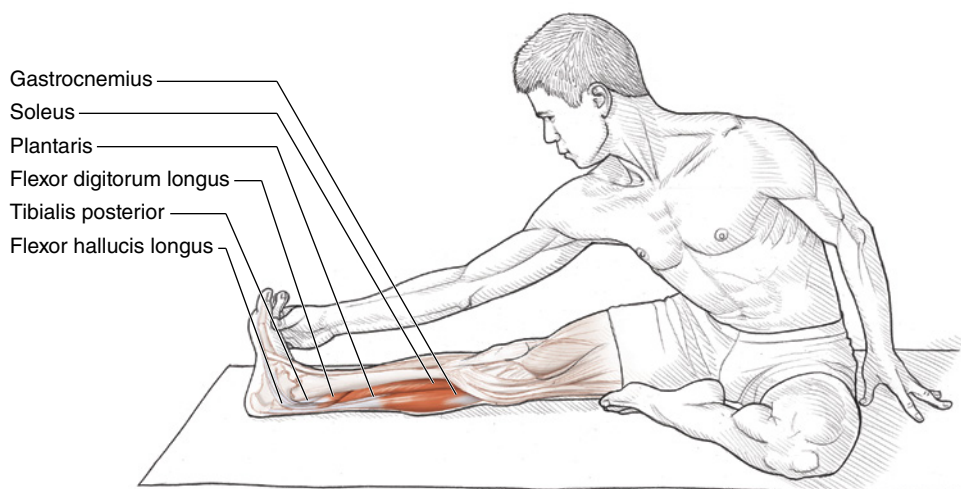
Bending the trunk toward the left knee instead of the right knee reduces the stretch of the most-stretched muscles on the left side of the body and increases the stretch of the most-stretched muscles on the right side of the body. Sitting with the right leg extended straight out in front, bend the left knee and place the left foot flat against the right inner thigh as close as possible to the pelvic area. Bend the trunk over toward the left (bent) knee as far as possible until you start feeling a slight stretch (light pain). Repeat on the opposite leg.



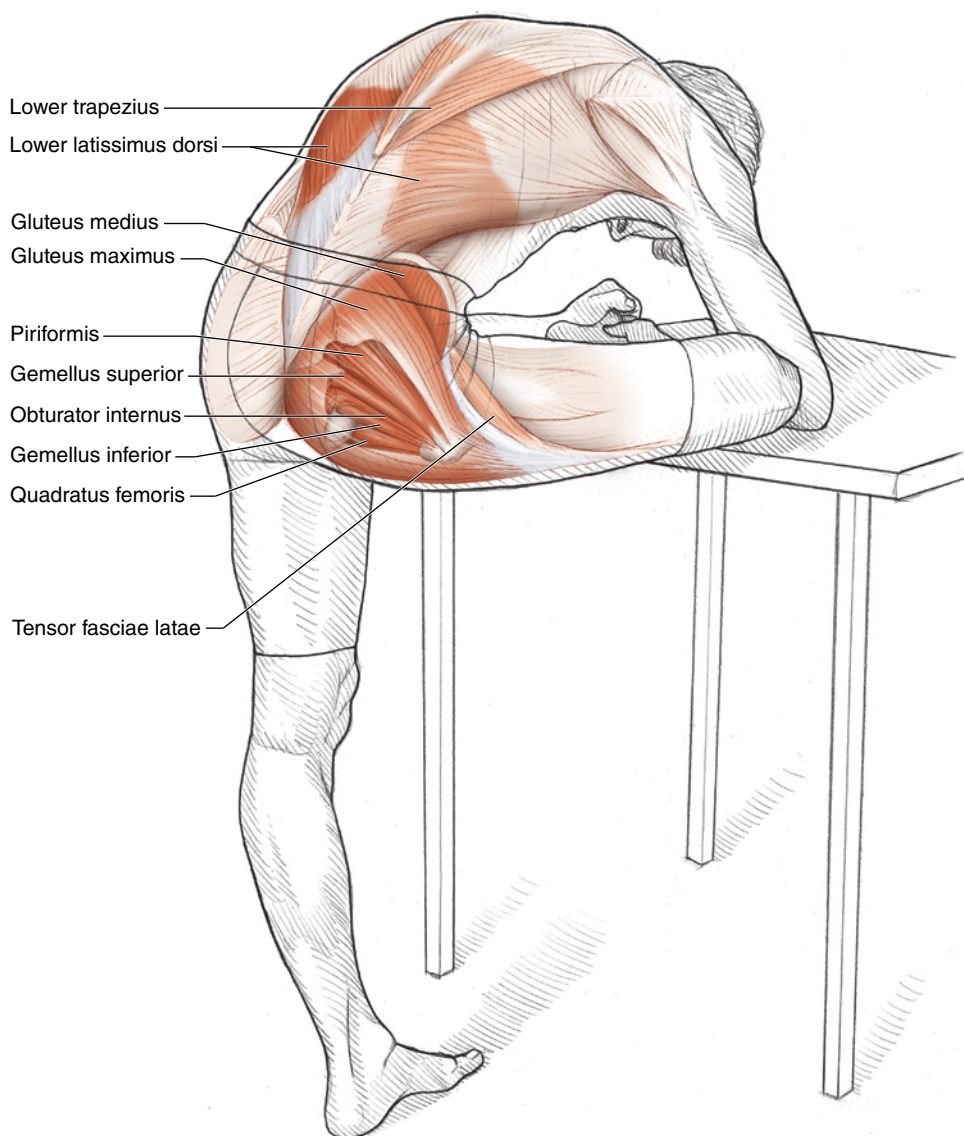
VARIATION

Intermediate Seated Hip External Rotator, Extensor, Knee Flexor, and Plantar Flexor Stretch

Modify the intermediate seated hip external rotator and extensor stretch to include the soleus, popliteus, flexor digitorum longus, flexor hallucis longus, tibialis posterior, gastrocnemius, and plantaris muscles of the lower leg as a combo stretch. Sitting with the right leg extended straight out in front, flex the left knee and place the left foot against the right inner thigh as close as possible to the pelvic area. Bend the trunk over toward the right (straight) knee as far as possible until you start feeling a slight stretch (light pain). As you bend forward, reach with the right arm, grasp the right foot, and pull the toes slowly toward the knee (dorsiflexed position).



ADVANCED STANDING HIP EXTERNAL ROTATOR STRETCH



Execution

1. Stand upright on the left leg, with the knee straight. Face a support surface such as a table, the edge of a couch, or a beam that is even with or just a little below the hips.
2. Bend the right leg at the hip at about a 90-degree angle and rest it on the support surface. The outside of the lower right leg rests as flat as possible on the surface. You can place a towel or pillow under the foot and lower right leg for cushioning.
3. Lower the trunk as far as possible toward the right foot, keeping the right knee as flat as possible on the surface.
4. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right gluteus maximus, right gluteus medius, right gluteus minimus, right piriformis, right gemellus superior, right gemellus inferior, right obturator externus, right obturator internus, right quadratus femoris, lower left erector spinae (iliocostalis, longissimus, spinalis), lower left latissimus dorsi

Less-stretched muscles: Right tensor fasciae latae, right lower latissimus dorsi, lower right trapezius

Stretch Notes

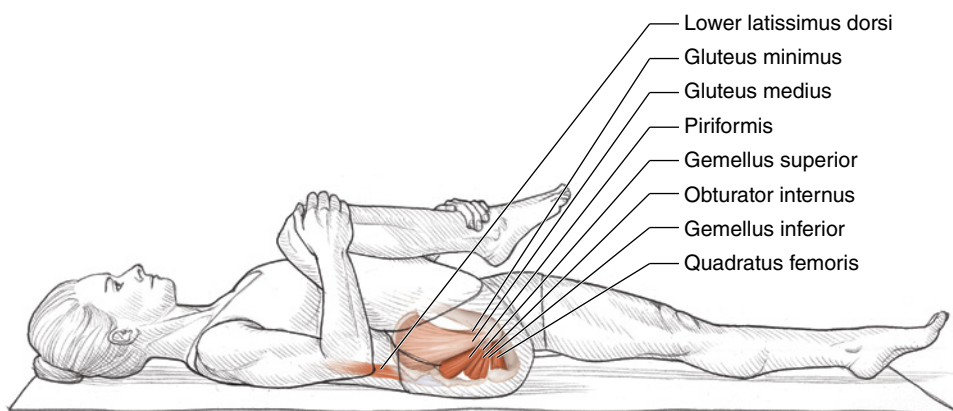
It is not uncommon to encounter periodic extensive soreness or tightness in the hip area as a result of certain types of exercise movements. Often this is due to extensive use of the hip external rotator muscles in activities such as ice skating, in-line skating, and the skating style of cross-country skiing. These muscles are located in the deep tissue of the hip just under the gluteus maximus muscle.

This is a more advanced stretch than the previous stretches in this chapter. It is one of the best stretches for the hip external rotator muscles. When placing the bent right leg on the supporting surface, make sure the entire lower leg is resting on it. This helps put the lower leg in a position of minimal stress on the knee joint. In addition, placing extra cushioning under the bent leg will make this stretch more comfortable.

Be sure to lower the trunk forward from the hip joint as far as you can. Keep the trunk as a straight unit; do not let the back curve. Bending the trunk toward the right knee instead of the left knee reduces the stretch of the most-stretched muscles on the right side of the body and increases the stretch of the most-stretched muscles on the left side of the body.

In addition, slowly adding more height to the right leg placement (perhaps a couple of inches after every two to four weeks) makes this stretch even more demanding. Increasing the height of the table, bench, or other surface up to 1 foot (30 cm) above the hips will increase the stretch to the highest possible level for these muscle groups.

RECUMBENT HIP EXTERNAL ROTATOR AND EXTENSOR STRETCH



Execution

1. Lie on your back on a comfortable surface.
2. While outwardly rotating the right leg, bend the right knee and bring the right foot to the body's midline. The knee is aligned outside of the chest and is pointed laterally. While keeping the left leg flat, grasp the right knee with the right hand and the right ankle with the left hand. Pull the lower leg as a unit as far as possible toward the chest.
3. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Right gluteus maximus, right piriformis, right gemellus superior, right gemellus inferior, right obturator externus, right obturator internus, right quadratus femoris, right lower latissimus dorsi, right erector spinae (iliocostalis, longissimus, spinalis)

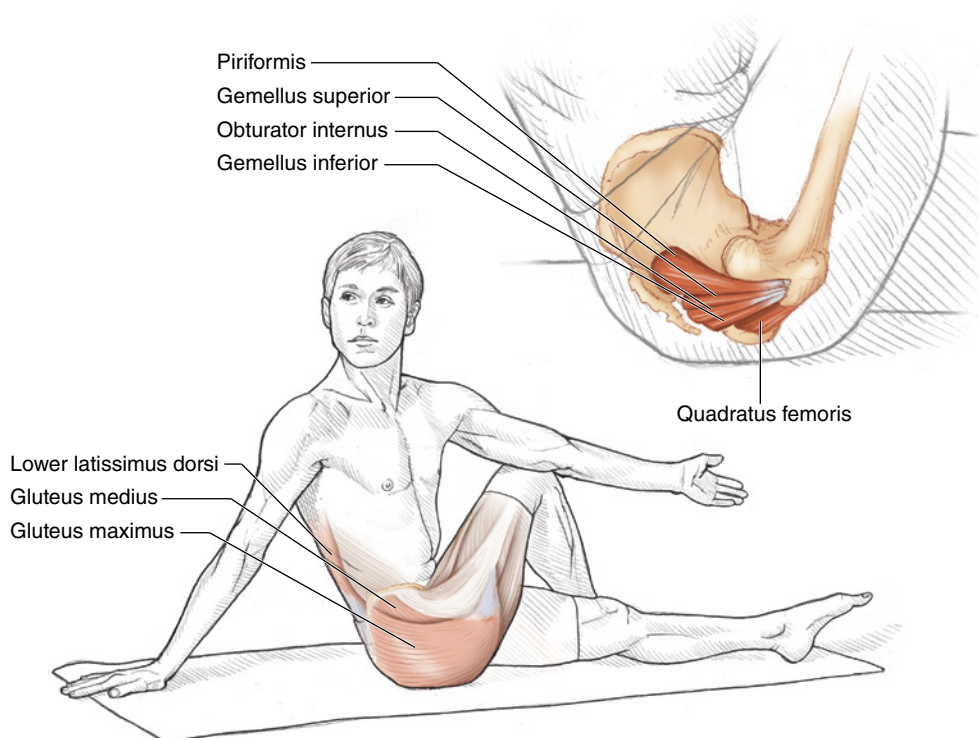
Less-stretched muscles: Right gluteus medius, right gluteus minimus

Stretch Notes

This is another version of a low-stress stretch for the hip external rotator and hip extensor muscles. These particular muscles can become sore or tight after engaging in activities that are not usual to daily routines or when unusual stress is placed on them. For instance, playing an impromptu game of soccer with your kids or friends where sprinting, jumping, and making sudden changes of direction are required can easily result in uncomfortable or painful muscles later on. There are also times when soreness is experienced, but it is difficult to recall what action or movement might have led to the aching muscles. In any case, when soreness or tightness is present, it is time to begin stretching the muscles affected. If you are new or relatively new to a stretching routine, this is a great stretch to begin with. As with many of the stretches in this book, it is easiest to begin a routine by sitting or lying down.

To maximize the effectiveness of this stretch, it is best to bring the ankle toward and over the head as far as possible. This will stretch the targeted muscles to the maximum level. Also, moving the ankle slightly to the right or left of the body will result in an additional pull on the multiple muscles of these hip rotators. Whenever you attempt a new or unaccustomed movement, as with the variations of this stretch, make sure to take safety precautions into consideration. In this case, put extra support behind the left knee with the left hand or a towel. In a bent position, such as in this stretch, the knee is vulnerable to injury, especially during experimentation with new movements.

HIP EXTERNAL ROTATOR AND BACK EXTENSOR STRETCH



Execution

1. Sit on the floor with the left leg extended.
2. Bend the right leg, and place the right foot on the outside of the left knee.
3. Bend the left arm, and position the outside of the left elbow against the outside of the upraised right knee.
4. Brace the right arm against the floor near the right hip.
5. Push the left elbow against the right knee, twisting the trunk as far as possible to the right. Maintain enough pressure with the left elbow to keep the right knee in a stable position.
6. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles on right side: Gluteus maximus, gluteus medius, gluteus minimus, piriformis, gemellus superior, gemellus inferior, obturator externus, obturator internus, quadratus femoris, lower latissimus dorsi, erector spinae (iliocostalis, longissimus, spinalis)

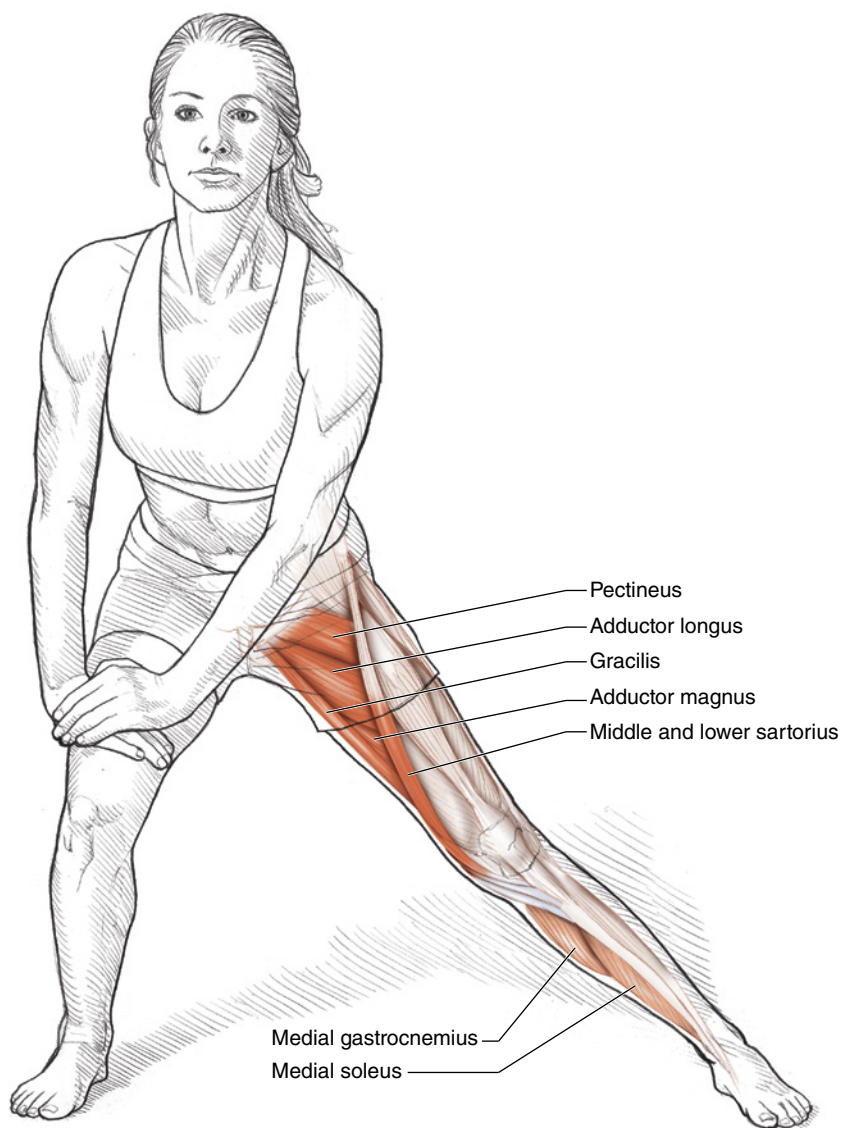
Less-stretched muscles on left side: Gluteus maximus, gluteus medius, erector spinae (iliocostalis, longissimus, spinalis), lower latissimus dorsi

Stretch Notes

This low-intensity stretch is well suited for those who have lower-back and hip pain. Lower-back problems can be quite common among any adult population but tend to become more prevalent as one ages. Pain in this area might be attributed to a specific injury or might just accumulate with use of the back muscles over time. Another cause of lower-back pain and discomfort is weakness of the back and abdominal muscles or muscular imbalance between these two muscle groups. This condition also tends to radiate pain sensations to the pelvic area, possibly limiting one's ability to comfortably accomplish daily tasks. To help alleviate this pain and discomfort, it would be beneficial to start performing this low-intensity stretch. Regular use of this stretch will strengthen this area and help reduce future reoccurrence of painful episodes.

While executing this stretch, try to keep the trunk upright. Do not arch the back or bend forward. Be careful to twist the trunk in a slow motion. This helps control the amount of stretch to the target muscles. Hold the position by bracing the right elbow against the left knee.

BEGINNER STANDING BENT-KNEE HIP ADDUCTOR STRETCH



Execution

1. Stand upright with the legs more than shoulder-width apart and the left foot turned out.
2. Lower the hips to a half-squatting position, bending the right knee and sliding the left foot out to the left to keep the left knee straight.
3. Place the hands on top of the right knee for support and balance, or hold on to an object for balance.
4. Repeat this stretch for the opposite leg.

Muscles Stretched

Most-stretched muscles: Left gracilis, left adductor magnus, left adductor longus, left adductor brevis, left pectineus, middle and lower left sartorius, left semitendinosus, left semimembranosus

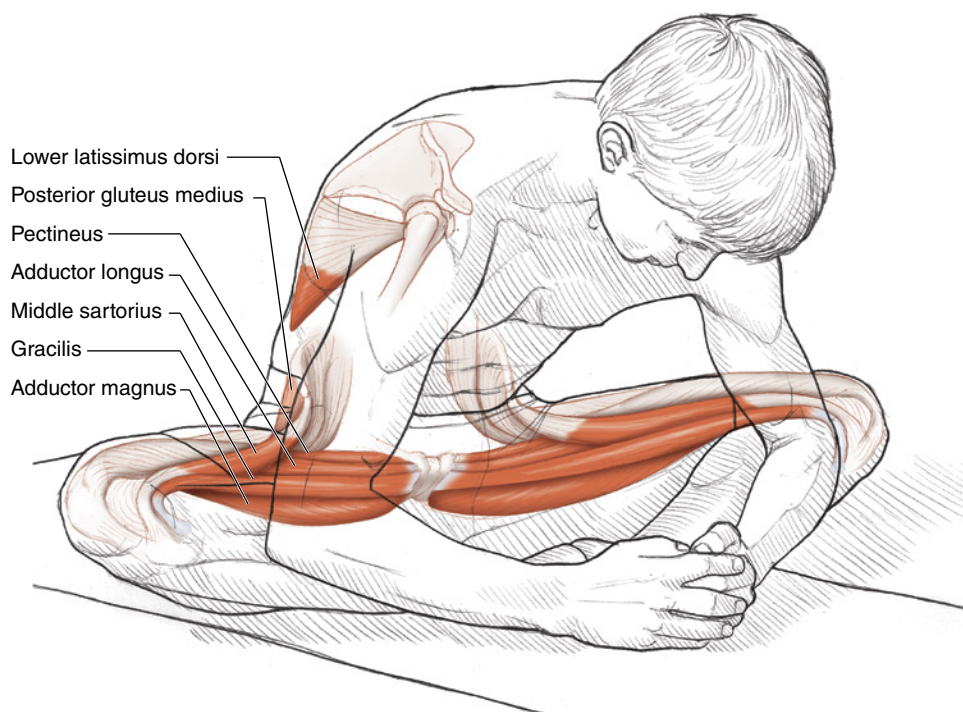
Less-stretched muscles: Medial left gastrocnemius, medial left soleus, left flexor digitorum longus

Stretch Notes

This is one of the easiest stretches for the inner thigh muscles. Most people do not use the inner thigh muscles to a great extent during normal daily activities. Consequently, these muscles tend to be weaker than other muscles in the thigh and hip areas and can become fatigued faster as a result. Participating in occasional activities such as walking or running on hilly terrain, climbing up and down stairs, or even playing a neighborhood basketball game with friends can sometimes lead to muscle twitching sensations, a sign of fatigue, in the inner thigh. If this occurs, it is recommended that the affected muscles be stretched for a couple of minutes to loosen them up. In most cases, the activity can then be resumed after stretching. It should be noted that it is always beneficial to perform a series of light stretches before starting any type of exercise, sport, or strenuous activity. This decreases the possibility of injury or discomfort to any muscle group.

While performing this stretch, keep the trunk as straight as possible. It is more comfortable to allow your weight to rest on the inside of the left foot. To increase the stretch, bend the trunk to the right, and press the right thigh down with both hands at the same time.

ADVANCED SEATED HIP ADDUCTOR STRETCH



Execution

1. Sit on the floor in the butterfly position—knees bent, feet together with the soles touching.
2. Bring the heels of the feet as close as possible to the buttocks. (Distance depends on the degree of your flexibility.)
3. Grasp the feet or just above the ankles, with the elbows spreading sideways and touching the legs just below the knees.
4. Bend the trunk over toward the feet, and press the lower part of the thighs and knees down with the elbows while stretching.

Muscles Stretched

Most-stretched muscles: Gracilis, adductor magnus, adductor longus, adductor brevis, pectineus, middle sartorius, lower erector spinae (iliocostalis, longissimus, spinalis), lower latissimus dorsi

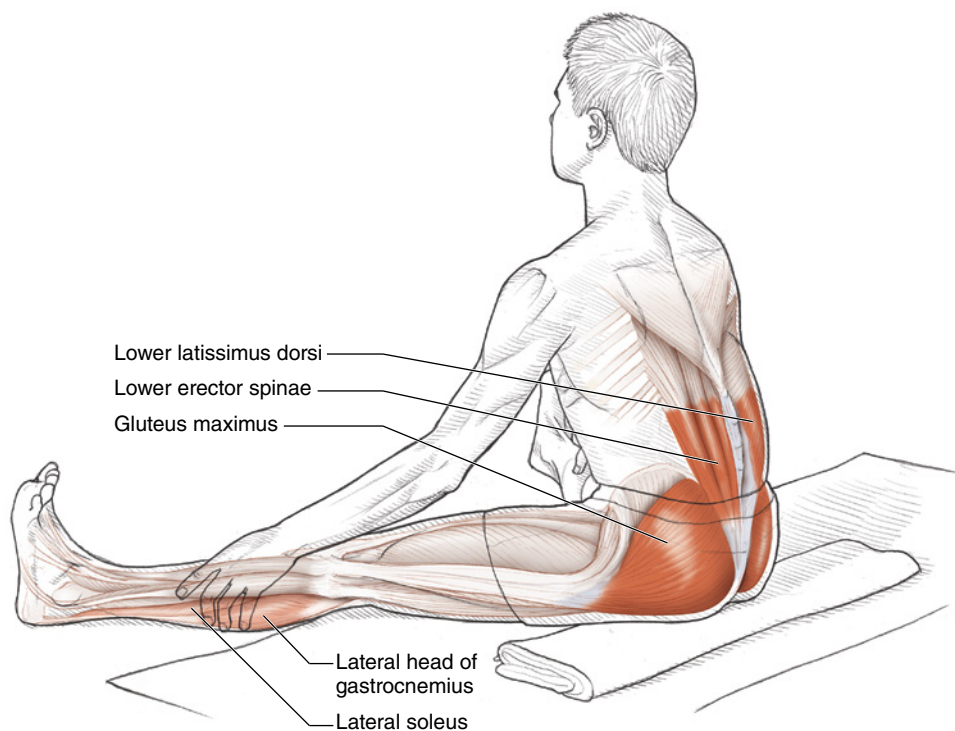
Less-stretched muscles: Gluteus maximus, posterior gluteus medius

Stretch Notes

The target muscles for this stretch—the adductor brevis, adductor longus, adductor magnus, gracilis, sartorius, and pectineus—are located on the medial (inner) side of the hip and thigh. These muscles are fairly large and are responsible for hip adduction (i.e., bringing the leg toward the midline of the body). Extensive use of hip adduction is typical in competitive or recreational activities such as ice skating, in-line skating, and the skating style of cross-country skiing. Most people engage in such activities on an occasional or seasonal basis. Unless training or conditioning is done as a regular routine, it is not uncommon for the more sporadic participant to encounter soreness or tightness after the activity. To prevent these symptoms from becoming more severe, it is recommended that these muscles be stretched before, during (if necessary), and after the activity.

The degree of stretch to the target muscles depends on the distance between the heels and the buttocks. The closer the heels are to the buttocks, the greater the stretch. In addition, the amount of stretch put on these adductor muscles can be controlled by the degree of pressure put on the lower part of the thighs and knees by the elbows. The stretch can be further intensified by grasping the feet and using them as a lever to pull the trunk forward. This technique not only targets the hip adductor muscles but acts as an effective stretch for the lower-back muscles as well. Placing the heels about 1 foot (30 cm) away from the buttocks increases the stretch on the gluteus maximus, gluteus medius, and erector spinae and places the greatest portion of the stretch on the origins of the adductor muscles.

SEATED HIP ADDUCTOR AND EXTENSOR STRETCH



Hamstrings:

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- This diagram shows a posterior view of the right leg, focusing on the hamstrings. The muscles are highlighted in red. Labels with leader lines point to the following muscles:
- Biceps femoris
 - Semitendinosus
 - Semimembranosus

Execution

1. Sit comfortably on the floor with legs extended in a V position, feet as far apart from each other as possible.
2. Place the hands on the floor next to the thighs.
3. Keep both knees straight and as flat against the floor as possible.
4. Slide the hands forward along the legs, and bend the trunk over between the knees.

Muscles Stretched

Most-stretched muscles: Semitendinosus, semimembranosus, gracilis, adductor magnus, adductor longus, gluteus maximus, lower erector spinae (iliocostalis lumborum, longissimus thoracis, spinalis thoracis), lower latissimus dorsi

Less-stretched muscles: Lateral soleus, lateral head of gastrocnemius, plantaris, biceps femoris

Stretch Notes

This is a more advanced stretch targeting the inside portion of the upper leg, the adductor muscles, as well as the inner backside of the thigh muscles, the semimembranosus and semitendinosus. In addition, it benefits the musculature of the lower back. Because of the nature of the position of this stretch, in which both legs are extended simultaneously, it is recommended for people who have already achieved a good amount of flexibility in this area of the body.

Keep both knees slightly bent while warming up. After the muscles are warmed up, you can move the knees into a straight position. To maximize the stretch, do not bend the knees, tilt the pelvis forward, or curve the back. Also, bend the trunk forward as a single unit, keeping it centered between the legs.

Changing the trunk position changes the nature of the stretch. For example, slowly moving the trunk over the right knee puts more stretch emphasis on the right-side hip extensors, right lower-back muscles, and left-leg adductor muscles. Conversely, moving the trunk to a position over the left knee emphasizes the stretch in the left-side hip extensors, left lower-back muscles, and right-leg adductor muscles.

(continued)

VARIATION

Seated Hip Adductor and Extensor Stretch With Toe Pull

By grasping the toes, you can make this stretch more complex and thus increase its effectiveness by including additional muscles. You can stretch not only the calf, hamstrings, posterior hip, lower back, shoulder, and arm muscles but also the entire right and left sides of the body at the same time. The amount of stretch depends on how hard you pull the toes toward the knees and the tibia bone. Simply execute steps 1 through 3 of the seated hip adductor and hip extensor stretch, and then for step 4 grasp the toes of both feet and pull them toward your head.

